

Document: Preprocessing Steps & Business Questions with Visualization

To ensure the final visual interface functions with absolute calculation accuracy and zero performance latency, a comprehensive data preprocessing framework was executed to bridge our backend data with Tableau. While the core dataset containing 4,416 rows of historical patient data was structurally sound, preparing it for multi-perspective clinical analysis required systematic metadata review, field formatting, and analytical calculation engineering.

1.Data Review & Exploration

connect to Heart_new2.csv to Tableau Public, will land on the **Data Source** tab.

1. **Verify Data Types:** Look at the icon above each column header to ensure Tableau has assigned the correct data type:
 - **# (Numeric / Decimal):** BMI
 - **# (Numeric / Whole):** PhysicalHealth, MentalHealth, SleepTime
 - **Abc (Text/String):** HeartDisease, Smoking, AlcoholDrinking, Stroke, DiffWalking, Sex, AgeCategory, Race, Diabetic, PhysicalActivity, GenHealth, Asthma, KidneyDisease, SkinCancer

NameHeart_new2.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remo...
Abc	Heart Disease Status	Heart_new2.csv	HeartD...
#	BMI	Heart_new2.csv	BMI
Abc	Smoking	Heart_new2.csv	Smoking
Abc	Alcohol Consumption	Heart_new2.csv	Alcohol...
Abc	Stroke	Heart_new2.csv	Stroke
#	Physical Health	Heart_new2.csv	Physic...
#	Mental Health	Heart_new2.csv	Mental...
Abc	Difficulty Walking	Heart_new2.csv	DiffWal...
Abc	Sex	Heart_new2.csv	Sex
Abc	Age Group	Heart_new2.csv	AgeCat...
Abc	Race	Heart_new2.csv	Race
Abc	Diabetic	Heart_new2.csv	Diabetic

2. Filtering and Structuring for Purpose

Depending on the business question, we may still need to filter the data to focus on specific subsets—such as certain time periods, regions, or product categories. Structuring the data to match the visualization goal helps ensure relevance and clarity.

3.Field Renaming & Final Formatting

To make your dashboards professional, rename columns to look clean and natural. You don't want technical names like DiffWalking appearing on your presentation charts.

1. **Rename Columns:** Double-click the column header text in the Data Source tab (or click the dropdown arrow on the field and select **Rename**). Change them to:
 - HeartDisease → **Heart Disease Status**
 - AlcoholDrinking → **Alcohol Consumption**

- DiffWalking → **Difficulty Walking**
- AgeCategory → **Age Group**
- GenHealth → **General Health Rating**
- SleepTime → **Sleep Duration (Hours)**
- PhysicalActivity → **Physical Activity**

Name Heart_new2.csv				
Description				
No description available.				
Fields				
Type	Field Name	Physical Table	Remo...	
Abc	Heart Disease Status ▼	Heart_new2.csv	HeartD...	
#	BMI ▼	Heart_new2.csv	BMI	
Abc	Smoking ▼	Heart_new2.csv	Smoking	
Abc	Alcohol Consumption ▼	Heart_new2.csv	Alcohol...	
Abc	Stroke ▼	Heart_new2.csv	Stroke	
#	Physical Health ▼	Heart_new2.csv	Physic...	
#	Mental Health ▼	Heart_new2.csv	Mental...	
Abc	Difficulty Walking ▼	Heart_new2.csv	DiffWal...	
Abc	Sex ▼	Heart_new2.csv	Sex	
Abc	Age Group ▼	Heart_new2.csv	AgeCat...	
Abc	Race ▼	Heart_new2.csv	Race	
Abc	Diabetic ▼	Heart_new2.csv	Diabetic	

4. Optional Calculated Fields

If needed, we can create calculated fields (e.g., profit margin, growth rate) to support deeper analysis. Even with a clean dataset, these additions can make our visualizations more insightful.

- created Heart Disease Case (Numeric), BMI Category, and High Risk Lifestyle Flag. This completely fulfills this requirement and adds deep clinical insight for your three scenarios.

Data	Analytics
Heart_new2	
Search	
Tables	
Abc Age Group	
Abc Alcohol Consumption	
Abc Asthma	
Abc BMI Category	
Abc Diabetic	
Abc Difficulty Walking	
Abc General Health Rating	
Abc Heart Disease Status	
Abc High Risk Lifestyle Flag	
Abc Kidney Disease	
Abc Physical Activity	
Abc Race	
Abc Sex	
Abc Skin Cancer	
Abc Smoking	
Abc Stroke	
Abc Measure Names	
Abc BMI	
Abc Heart Disease Case (Numer...	
Abc Mental Health	
Abc Physical Health	
Abc Sleep Duration (Hours)	
Abc Heart_new2.csv (Count)	
Abc Measure Values	

5. Validation for Accuracy

Lastly, a quick validation against the source or summary metrics ensures everything is accurate. This final step helps maintain trust in the insights generated.

2. Target Business Questions & Stakeholder Requirements

The design configuration of our analytical dashboard is intentionally constructed to answer three primary operational inquiries, matching the direct business needs of our target personas:

Question 1 (Clinical Diagnostic View — Dr. Sharma)

"What are the primary high-risk patient lifestyle groups isolated by localized age brackets, BMI classifications, and active tobacco smoking habits?"

- **Analytical Target:** Eliminates the need for slow, manual table reading by allowing clinicians to perform 1-click filtering to rapidly identify high-priority target patient segments requiring proactive therapeutic or preventative intervention plans.

Question 2 (Epidemiological Policy View — Ramesh)

"How do localized public health metrics and cardiac incident trends split across urban versus rural demographic sectors?"

- **Analytical Target:** Empowers public health program administrators with explicit, visual geographic evidence required to back up regional resource distribution, prioritize community budgets, and drive healthcare policy deployments.

Question 3 (Patient Benchmark View — Anita)

"How can an individual patient's personal health parameters be dynamically cross-referenced and tracked against official healthcare guidelines?"

- **Analytical Target:** Lowers patient anxiety and promotes health awareness by translating intimidating, raw data tables into interactive visual markers that place personal clinical statuses safely within context against standard target benchmarks.

3. Exploratory Visualizations & Verification

(Use this section to showcase your preliminary findings or verification graphics before you export your final file to PDF.)

Visual Element A: Clinical Cohort Screening View

- **Functional Description:** Explores the physical correlations between escalating body mass variables, aging brackets, and systemic heart disease presence.

Visual Element B: Demographic Spread & Location Analysis

- **Functional Description:** Compares the structural division of historical case counts categorized cleanly by regional and demographic indicators.

